HUISEOK MOON

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RESEARCH INTERESTS

Assistive and wearable robotics, physical human–robot interaction, multimodal human-motion and intent inference, and nonlinear/model-based/adaptive control. Current research integrates physics-based modeling and simulation with real-time machine learning and multimodal sensing to understand, predict, and assist human motion.

Keywords: Wearable/Assistive Robotics • Machine Learning • Multimodal Sensor Integration • Nonlinear Control

EDUCATION

Université Paris-Est Créteil — **Créteil, France**
Ph.D. in Signal, Image and Control — Advisor: Dr. Samer Mohammed — *Nov 2019 – Jan 2024*
Thesis: *Human-in-the-loop Paradigms for Mobility Assistance Using Powered Exoskeleton*

Sogang University — **Seoul, South Korea**
M.S. in Mechanical Engineering — Advisor: Dr. Kyoungchul Kong — *Mar 2017 – Feb 2019*
Thesis: *Control Algorithm of a Robotic Leg for Reducing Interaction Force Between a Load and the Human Body*
Graduate Research Fellowship, Ilwun Science and Technology Foundation (2017–2019)

Kwangwoon University — **Seoul, South Korea**
B.S. in Robotics (Intelligence System) — *Mar 2010 – Feb 2017*
Thesis: *Research on Personal Mobility Using Ball Balancing Robot*

RESEARCH EXPERIENCE

Postdoctoral Researcher — **Jul 2026 – Present**
Université Paris-Est Créteil — *Vitry-sur-Seine, France*

- Designed a framework for **human movement representation** and **real-time joint moment estimation**, learning shared patterns across walking conditions and using wearable sensor data to estimate ankle moments.
- Developed a **multi-IMU gait assessment system** and conducted experiments involving people with **post-stroke hemiparesis** to evaluate lower-limb kinematics and measurement reliability across walking conditions.

Postdoctoral Researcher — **Sep 2025 – Mar 2026**
Korea Advanced Institute of Science and Technology (KAIST) — *Daejeon, South Korea*

- Investigated **physics-informed** human movement analysis and **representation learning** using a multimodal biomechanics dataset.

Research Engineer — **Sep 2023 – Aug 2025**
Université Paris-Est Créteil — *Vitry-sur-Seine, France*

- Developed **LSTM-based gait recognition** and adaptive ankle assistance using two foot-mounted IMUs, achieving **98% mean recognition accuracy** across five gait modes during assisted walking in healthy participants.
- Developed **adaptive FES for foot-drop correction** in patients, using **EMG signals** to analyze muscle dynamics and adapting stimulation timing and intensity in real time for energy-efficient assistance.
- Analyzed IMU signals from a portable alternometer to quantify forearm pronation–supination for **objective assessment of parkinsonian motor impairment**.

ATER (Teaching and Research Assistant) — **Nov 2022 – Aug 2023**
Université Paris-Est Créteil — *Vitry-sur-Seine, France*

- Led **undergraduate laboratory sessions (TP)** on wearable sensor integration and biomechanical signal processing using Python and MATLAB.

Research Assistant — **Nov 2019 – Oct 2022**
Université Paris-Est Créteil — *Vitry-sur-Seine, France*

- Developed nonlinear control strategies for lower-limb robotic assistance, including **impedance modulation** for sit-to-stand movements and **adaptive fuzzy control and model predictive control** for reference tracking in actuated ankle and knee orthoses during walking. Integrated IMU, EMG, and FSR data to adjust assistance based on user states and task demands.
- Combined **online random-forest gait classification** with task-specific ankle torque assistance across level walking, stair negotiation, and ramp traversal.

Research Assistant — **Mar 2017 – Feb 2019**
Sogang University — *Seoul, South Korea*

- Developed **full lower-limb exoskeleton control** to reduce interaction forces between the load and the body in high-mobility robotics, analyzed human gait using **OpenSim and Vicon**, and participated in a quadruped robot project.

PUBLICATIONS

Journals

2026 X. Luo*, **H. Moon***†, J.-M. Gracies, W. Huo†, and S. Mohammed†. Knee-Guided Phase–Magnitude Decoupled Adaptive Functional Electrical Stimulation for Drop-Foot Correction. Submitted to *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 2026. (Under Review) * Equal contribution; † Corresponding author.

2026 L. Marusu, C. Gault-Colas, **H. Moon**, S. Mohammed, and J.-M. Gracies. Changes in Walking Speed, Step Length, and Lower-Limb Kinematics over Eight Weeks of Guided Self-Rehabilitation Contracts in Hemiparetic Patients: A Prospective Cohort Study. Submitted to *Journal of Neural Engineering*, 2026. (Under Review)

2026 O. Bey, M. Chemachema, R. Jradi, **H. Moon**, H. Rifai, Y. Amirat, and S. Mohammed. Non-Model-Based Finite-Time Adaptive Neural Output Feedback Control for an Active Ankle–Foot Orthosis. *IEEE Transactions on Automation Science and Engineering*, vol. 23, pp. 14103–14119, 2026. 10.1109/TASE.2026.3718534

2025 **H. Moon**, O. Bey, A. Boubezoul, L. Oukhellou, and S. Mohammed. Real-Time LSTM-Driven Dynamic Gait Mode Detection for Enhanced Control of Actuated Ankle–Foot Orthosis. *IEEE Transactions on Robotics*, vol. 41, pp. 4794–4809, 2025. 10.1109/TRO.2025.3593111

2023 G. Khodabandelou, **H. Moon**, Y. Amirat, and S. Mohammed. A Fuzzy Convolutional Attention-Based GRU Network for Human Activity Recognition. *Engineering Applications of Artificial Intelligence*, vol. 118, Art. no. 105702, 2023. 10.1016/j.engappai.2022.105702

2022 **H. Moon**, R. Maiti, K. Das Sharma, Y. Amirat, P. Siarry, and S. Mohammed. Hybrid Half-Gaussian Selectively Adaptive Fuzzy Control of an Actuated Ankle–Foot Orthosis. *IEEE Robotics and Automation Letters*, vol. 7, no. 4, pp. 9635–9642, 2022. Presented at ICRA 2023 (oral presentation). 10.1109/LRA.2022.3191187

2022 I. Jammeli, A. Chemori, **H. Moon**, S. Elloumi, and S. Mohammed. An Assistive Explicit Model Predictive Control Framework for a Knee Rehabilitation Exoskeleton. *IEEE/ASME Transactions on Mechatronics*, vol. 27, no. 5, pp. 3636–3647, 2022. 10.1109/TMECH.2021.3126674

2022 W. Huo, **H. Moon**, M. A. Alouane, V. Bonnet, J. Huang, Y. Amirat, R. Vaidyanathan, and S. Mohammed. Impedance Modulation Control of a Lower-Limb Exoskeleton to Assist Sit-to-Stand Movements. *IEEE Transactions on Robotics*, vol. 38, no. 2, pp. 1230–1249, 2022. 10.1109/TRO.2021.3104244

Conference Proceedings

2024 O. Bey, R. Jradi, **H. Moon**, H. Rifai, K. Das Sharma, Y. Amirat, and S. Mohammed. A Novel Funnel-Based L_1 Adaptive Fuzzy Approach for the Control of an Actuated Ankle Foot Orthosis. *IEEE International Conference on Robotics and Automation (ICRA)*, pp. 17487–17493, 2024. 10.1109/ICRA57147.2024.10610584

2022 **H. Moon**, A. Boubezoul, L. Oukhellou, Y. Amirat, and S. Mohammed. Online Human Intention Detection through Machine-Learning Based Algorithm for the Control of Lower-Limbs Wearable Robot. *IEEE-RAS International Conference on Humanoid Robots (Humanoids)*, pp. 809–814, 2022. 10.1109/Humanoids53995.2022.10000150

2021 P.-G. Jung, W. Huo, **H. Moon**, Y. Amirat, and S. Mohammed. A Novel Gait Phase Detection Algorithm for Foot Drop Correction through Optimal Hybrid FES–Orthosis Assistance. *IEEE International Conference on Robotics and Automation (ICRA)*, pp. 10391–10397, 2021. 10.1109/ICRA48506.2021.9561497

ACADEMIC SERVICE

Peer Review	IEEE Transactions on Robotics, IROS, ICRA, Engineering Applications of Artificial Intelligence, Mechatronics, and others
Invited Talks	REHAssist Visiting Seminar, EPFL, Switzerland (2025); LISSI–BIOTN Joint Seminar, UPEC, France (2024); LISSI Research Seminars, UPEC, France, 4 sessions (2020–2023)
Volunteering	IFAC World Congress, volunteer (2025); IUT Research Day, poster exhibition (2024); Fête de la Science, public outreach exhibitor (2023); Korea Defense Science & Tech Fair, exhibitor (2017)
Mentoring	M.S. students: L. Marusu (2025), A. A. Benhamlaoui (2024), D. Plet (2024), W. Legendre (2023), J. M. Fadous (2022), R. Ravaka & M. Diaby (2021). Topics spanning multi-sensor gait analysis, AI-based gait event detection, Parkinson's diagnostics, and assistive robot control

SKILLS

Programming	Python, C, C++, MATLAB, LabVIEW, SolidWorks, LaTex
Frameworks	PyTorch, TensorFlow, Simulink
Expertise	Real-time ML deployment, Multimodal sensor fusion, Embedded systems, Nonlinear control, Human–robot interaction, Wearable robotics, Signal processing
Languages	Korean (native), English (fluent), French (basic)

REFERENCES

Dr. Samer Mohammed
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